What is a *life cycle?*

Birds, and most reptiles and amphibians, lay fertilized eggs that develop outside the mother's body.

Life is always changing, but it follows patterns too. Living things—including ourselves—grow, produce young, and die. There are repeating processes for nonliving things too, from mountains, rocks, and rivers to planets, comets, and stars. We call all these patterns life cycles.

Life cycles are interlinked. Plants take nutrients and water from the soil and energy from sunlight. Animals eat plants or other animals in order to grow. Many plants rely on animals such as insects to spread their pollen so they can make seeds and reproduce. When plants and animals die, their remains rot and become part of the soil that will nourish new plants.

The life cycles we see around us appear endless. Birth and growth are balanced by breakdown and decay. Yet these cycles can be fragile. Change—whether natural or caused by humans—can disrupt them. Vulnerable species may dwindle to extinction when their cycles get broken, while—over millions of years—new species emerge by evolution.

Earth and space

Our Earth is constantly changing. Rocks wear away and get recycled into new rocks, while water circulates between sea, sky, and land. In space, comets and stars form from gas and dust, burn out, then return to gas and dust.

***Earth's life cycles** are tiny fractions of the age of the universe: 13.8 billion years.*